

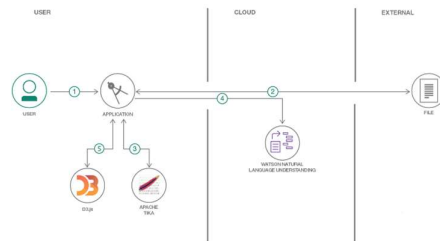
Project Design Phase-II
Data Flow Diagram & User Stories

Date	03 April 2026
Team ID	SWTID-2026-4754
Project Name	Automation Employee Data
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

1. Functional Requirements

Functional requirements describe what the system should do. They define the core features and operations of the system.

FR-1: User Registration

The system should allow users to create an account using multiple methods:

- Registration through a form by entering personal details like name, email, and password
- Registration through Gmail integration for quick signup

- Registration through LinkedIn for professional profile-based signup

FR-2: User Confirmation

After registration, the system must verify user identity:

- Confirmation via email with a verification link
- Confirmation via OTP (One-Time Password) sent to the registered email or mobile number

FR-3: User Login

The system should allow registered users to log in securely:

- Login using email and password
- Login using Gmail or LinkedIn
- Password recovery option using email

FR-4: Employee Data Management

The system should allow users/admins to manage employee records:

- Add new employee details
- Update existing employee information

- Delete employee records
- Search and view employee details

2. Non-Functional Requirements

Non-functional requirements define how the system performs and its quality attributes.

NFR-1: Usability

The system should be user-friendly and easy to navigate:

- Simple interface design
- Clear instructions and labels
- Accessible for all users

NFR-2: Security

The system must protect user data:

- Data encryption for sensitive information
- Secure login authentication
- Role-based access control

NFR-3: Reliability

The system should work consistently without failure:

- Minimal system crashes
- Accurate data processing
- Proper error handling

NFR-4: Performance

The system should respond quickly:

- Fast loading time
- Efficient data processing
- Support multiple users simultaneously

NFR-5: Availability

The system should be available at all times:

- 24/7 accessibility
- Minimal downtime

- Backup and recovery support

NFR-6: Scalability

The system should handle growth:

- Ability to support increasing users
- Efficient handling of large data
- Flexible architecture for expansion

Conclusion

Functional requirements define what the system does, while non-functional requirements ensure the system performs efficiently, securely, and reliably. Together, they help in designing a high-quality and user-friendly solution.